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1. What is data Science? Explain its primary objective

Data Science is the Field of using scientific methods, stastics, programming and machine learning to extract useful information from data.

Objective: To analyze data and generate meaningful insights that support better decision-making.

3. List any four benefits of data science in Buisness and society.

1. Helps in Better Decision Making
2. Improves buiseness efficiency and productivity
3. Enables prediction of future trends.
4. Helps in solving real-world problems such as healthcare and traffic management.

5. What are the different facets of data? Briefly explain each.

The main facets of data are:

1. Structured data - Organised in rows and columns, such as database tabels.
2. semi-structured data - Has some organization but no fixed table structure, such as JSON and XML.
3. unstructured data - Has no predefined structure, such as images, videos and text.

7. What is Big Data ? State its key characteristics (5V's)

Big data refers to extremely large and complex datasets that are difficult to process using traditional methods.

The 5V's are:

1. Volume: Huge amount of data
2. Velocity: Speed at which data is generated.
3. Variety: Different types and formats of data.
4. Veracity: Accuracy and reliability of data.

9. What are the major phases of the Data science process?

The major phases are:

1. Data collection
2. Data cleaning and preparation
3. Data analysis and exploration
4. Model building
5. Mode evaluation and deployment

11. Why is data collection considered an important step in the Data Science process?

Data collection is important because the quality and quantity of data directly affect the accuracy of analysis and models. Good quality data helps produce reliable insights and predictions.

13. What is data integration ? Why is it necessary in Data Science. ?

Data integration is the process or combining data from different sources into a single, unified dataset

it is necessary because it provides a complete view of the information and makes analysis more accurate and useful.

15. What is data Analysis ? Mention any four techniques used for data Analysis.

Data Analysis is the process of examining and interpreting data to discover useful patterns and insights.
Four techniques are:

1. stastical Analysis
2. correlation Analysis
3. Regression Analysis
4. clustering

17. What is Meant by Building a model in data science ? Mention the Basic steps involved.

Model Building is the process of creating a mathematical or machine-learning model that learns patterns from data.

Basic steps:

1. Select an appropriate algorithm.
2. Train the model using data.
3. Test the model
4. Evaluate and improves its performance.

9. Why is processing findings an essential part of the data science process ? Mention any four visualization

tools or techniques.

presenting Findings helps communicate complex data and insight clearly so that users can understand them and make decisions.

Four visualization tools/techniques are:

1. Matplotlib
2. Seaborn
3. Tableu
4. power BI